Trade Study: Vehicle Options

- **Conducted by:** Heather Cassel
- **Decision:** GEEK Smart Robot

Problem: A vehicle needs to be chosen that can be used with the NVIDIA Jetson Nano in order to create a vehicle capable of moving autonomously while tracking an object. Options vary from dedicated AI robotics kits designed to work with the Jetson Nano to hobbyist-level RC cars with minor modifications via an external driver board to gain control of the vehicle.

		JetRacer AI Kit	GEEK Smart Robot	Mostop RC w/ Driver Board
Criteria	**Weights**	**Grade**	**Grade**	**Grade**
Setup Time	0.20	9	7	2
Price	0.15	2	2	9
Surface Area	0.15	5	9	3
Payload Capacity	0.20	5	10	0
Kinematics	0.10	9	7	8
Software Ecosystem	0.20	10	9	8
Average Score		40/60	44/60	30/60
Weighted Score		6.75	7.55	4.60

Breakdown

1st Choice: GEEK Smart Robot

With the highest overall rating of 7.55, the ideal option for this project would be the GEEK Smart Robot platform. While this option is the most expensive of the three at $229, it proves its worth by excelling in all other criteria. This platform is designed to work with developer boards like the Jetson Nano, making the setup quick and intuitive. The vehicle design provides

a large, flat surface with various holes and slits for ample room and easy attachment of accessories. With a payload capacity of 5 kg (11 lbs), this platform would have no issue carrying and running as intended with the addition of the camera and sensors we will be adding. The omni-directional steering, while not realistic to most ground operated vehicles, allows the robot to turn on a dime and could be a good option to simulate the turning capability of air travelling vehicles. Finally, this vehicle is engineered to be usable with a variety of open-source robotics ecosystems, such as the ROS / ROS2 ecosystem which utilizes the Linux environment of the NVIDIA Jetson Nano.

2nd Choice: JetRacer AI Kit

The JetRacer AI Kit is a strong second contender. This platform is specifically designed to work with the NVIDIA Jetson Nano. With a pricepoint of $214, it is similar in price to the GEEK Smart Robot, however it falls behind in terms of surface area and payload capacity. This model does come included with a front facing camera, and the kit itself is designed to be a self-driving, road tracking vehicle, but since it is built with a specific purpose in mind the available space to add more sensors like our LiDAR is limited, and the weight capacity decreases to about 1.5-2.5 kg (3-5 lbs). This weight capacity should still be sufficient for our purposes, but does run a higher risk of affecting the vehicle's performance once it is fully loaded. This vehicle utilizes an ackerman steering setup, which would more closely resemble a land operated vehicle when driving. The existing software ecosystem is designed to work with the Jetson Nano, including projects like DonkeyCar and JetRacer, which work with NVIDIA, Tensorflow, OpenCV, Python, PyTorch, and more.

3rd Choice: RC Car + Driver Board Combo (not recommended)

The RC Car and Driver Board combination is possible, but not recommended. The setup would take significantly longer since the RC car is not natively designed to work with a computer. Once setup, the vehicle itself also presents some severe limitations. While the price is low at about $65 for the vehicle and board, the surface area and payload capacity make this vehicle nearly impossible to use. The plastic body of the vehicle would have to be completely removed to make room for the driver board, computer, camera, and sensors, and with the combined weight of these materials, they will likely be heavier than the truck itself and risk rendering the vehicle immobile by bottoming out the suspension. The vehicle does have the benefit of realistic driving when compared to typical land driving vehicles, and can leverage the same open-source ecosystems as the GEEK (including ROS / ROS2). Overall, this option is theoretically possible but would prove to be extremely difficult and would require a much longer startup time.